ASSIGNMENT 2

Apply Clustering Techniques Using Unsupervised Learning

1. Aim

To apply the K-Means clustering algorithm to group similar data points using unsupervised learning.

2. Objectives

1)To understand the concept of unsupervised learning

2)To implement the K-Means clustering algorithm.

3)To understand clustering and its importance.

4)To identify groups of similar data points.

5)To visualise clusters and their centers.

.3. Theory

Machine learning is commonly divided into supervised learning and unsupervised learning.

In supervised learning, a model learns from data containing known output labels. In unsupervised learning, the data do not contain supplied target labels, and the algorithm attempts to discover patterns or structures.

Clustering is an unsupervised learning technique used to divide data into groups called clusters. Data points within a cluster should be relatively similar according to the selected features and distance measure.

K-Means is one of the most widely used clustering algorithms. The value of K represents the number of clusters to create.

The K-Means algorithm works as follows:

1.Select the required number of clusters.

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3.Assign each data point to its nearest cluster centre.

4.Calculate new centres by taking the mean of the points assigned to each cluster.

5.Repeat the assignment and centre-update steps until the centres or assignments stabilise.

Feature scaling is often necessary because K-Means relies on distances. If one feature has a much larger numerical range than another, it may dominate the distance calculation.

4. Algorithm

- Standardise the features.
- Create a K-Means model with three clusters.
- Fit the model and obtain cluster labels.
- Display the cluster labels and centres.
- Visualise the clusters using a scatter plot.

5. Explanation of Important Code

- make_blobs() generates sample data arranged around three groups.
- StandardScaler() standardises the features.
- KMeans(n_clusters=3) creates a model that aims to form three clusters.
- fit_predict() trains the model and returns a cluster label for each point.
- cluster_centers_ provides the final coordinates of the cluster centres.

- plt.scatter() displays the points and their assigned clusters.
- The X-shaped markers represent the cluster centres.

6. Expected Output

The terminal displays the first 15 cluster labels and the coordinates of the cluster centres.

A scatter plot appears with three colour-coded clusters. X-shaped markers indicate the centres of the clusters.

7. Applications

I. Customer segmentation in marketing.

II. Grouping similar documents and news articles.

III. Analysing patterns in sensor measurements.

IV. Grouping products according to their characteristics.

V. Exploring patterns in geographical data.

VI. Supporting image segmentation tasks.

8. Advantages

1. K-Means is relatively simple to understand and implement.
2. It does not require pre-labelled classes.
3. It can process many numerical datasets efficiently.
4. Cluster centres provide a useful summary of each group.

9. Limitations

1. The number of clusters must usually be selected in advance.
2. Results may depend on initial cluster centres.
3. Outliers can influence the cluster centres.
4. It may not perform well when clusters have irregular shapes or very different densities.

10. Conclusion

The K-Means clustering algorithm was implemented to divide sample data into three groups. The resulting cluster labels and centres were visualised using a scatter plot. The experiment demonstrated how unsupervised learning can discover patterns in data without predefined class labels.